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Exploring the value of CO_2 shipping transport flexibility in international batchwise transportation carbon capture and storage networks

Denis Martins Fraga[a*], Anna Korre[a,b], Zhenggang Nie[a], Şevket Durucan[a]

[a]*Department of Earth Science and Engineering, Royal School of Mines, Imperial College London, London SW7 2AZ, United Kingdom*
[b]*Energy Futures Lab, Imperial College London, London SW7 2AZ, United Kingdom*

Abstract

The benefits of CO_2 shipping transport within international Carbon Capture and Storage (CCS) networks have been growing in importance as it offers flexibility and allows to postpone investment in capital-intensive infrastructure until enough CCS projects are launched. This work explores this value through a comparative cost assessment of 15 what-if scenarios devised for standalone and integrated CCS networks, designed around the Northern Lights project and the UK's Teesside cluster. The analysis is performed by implementing a Multi-period, Multi-mode, and Multi-objective Mixed integer linear programming (MoMilp) model developed at Imperial College London and considering all the elements of the CCS value chain after CO_2 is captured (i.e. conditioning, intermediate storage and injection in the saline aquifer), allowing to account for individual regional characteristics in terms of design and network costs. The configurations of the different scenarios are used to analyse the implications of CCS network development hold up, e.g. for a part of the network to become available at a later period, or for the CO_2 emission growth to occur at a later time in the planning horizon. The simulation results illustrate the interplay between postponing CCS network infrastructure investment and generating gains of scale in an existing network. This flexibility between CCS networks could provide relevant cost reduction while maximising the storage of CO_2.

Keywords: Batchwise transportation; Carbon Capture and storage (CCS) networks; CCS network integration

1. Introduction

The importance of carbon capture and storage (CCS) in Europe is growing, with a 61% increase in commercial-scale projects, reaching 119 projects in 2023. The main driver for this growth is the support secured from the European Commission and national governments, including funding mechanisms (i.e. Innovation Fund, Horizon Europe, and Cluster Sequencing in the UK), large-scale development plans (e.g., Green Deal Industrial Plan) and legislative support from both the European Commission and the UK [1, 2]. The increasing maturity of these projects starts shaping the infrastructure available for storing increasing amounts of CO_2, helping countries achieve their decarbonisation targets.

The most advanced project (i.e. Northern Lights) is characterised by batchwise transportation, facilitated by CO_2 transport ships, and injection via a relatively short pipeline to storage formations in the Norwegian North Sea. The transport infrastructure is supported by two intermediate storage hubs that aim to help manage and assure the flow in the network. Multiple other opportunities are being studied in different regions to test multi-mode transportation options. However, such efforts are at different stages of maturity and are often developed considering an optimum

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country-focused design without accounting for adjacent CCS networks' development due to the monopolistic nature of such first-of-a-kind infrastructure.

This work analyses and presents the value of batchwise CCS transport networks' flexibility by performing comparative cost assessments of standalone CCS networks with scenarios identified to serve what-if analysis and to assess the benefits of integrated CCS network design. The what-if analysis conceptualisation is applied for batchwise CCS network analogues of the Nothern Lights project (NL), the Stella Maris concept (SM), and the Teesside cluster in the UK, as this is the most advanced track-1 project.

2. Methodology

The approach follows the 4-step methodology described in Fig. 1. The first step consists of the cost data gathering and the bottom-up cost design. For the batch-wise infrastructure (e.g., vessels, intermediate storage and conditioning), the costs are sourced from existing models developed by the authors [3,4,5] and account for the regional cost differences between the networks, considering ships with 6, 20, and 50 CO_2 kt CO_2 capacity, three onshore (i.e. 6", 12", 18") and offshore (i.e. 36", 40", 48") pipeline diameters.

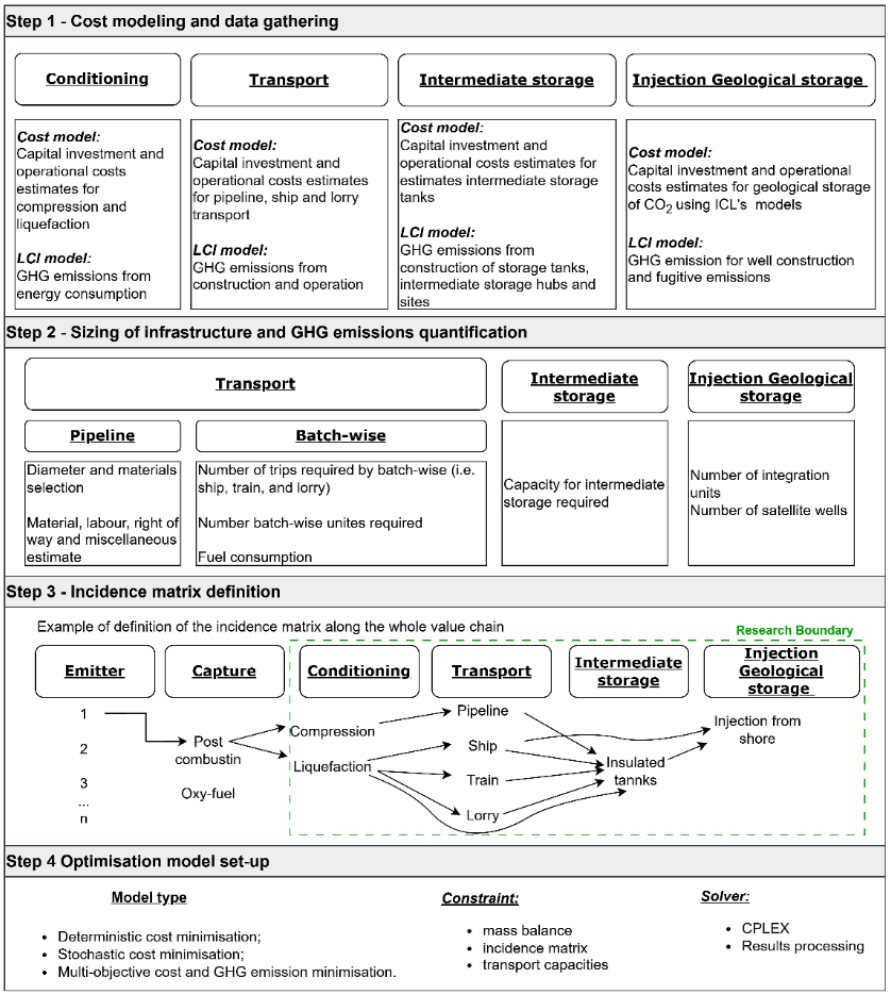


Fig. 1. Cost and emissions accounting, scenario analysis and optimisation methodology implemented.

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The bottom-up cost models devised for geological storage sites in [6] were also considered for different field development strategies (i.e. a platform with wells for the UK and subsea design for NL and SM). The design of the intermediate storage site considers one cargo of ship from the highest upstream ship travelling to the facility.

The remaining three steps follow the models devised in [3,4,5,7,8], considering only deterministic multi-period (i.e. 30 years in a yearly resolution) network cost minimisation according to equation (1), where $TC_{(k,i,j,t)}$ is the total cost (€) from transport mode k, between locations i and j, in period t. The total cost comprises conditioning (i.e., liquefaction and compression), intermediate, transport and injection costs. Stochasticity and multi-objective optimisation were not considered in this implementation, although the models developed by the authors can account for this in [5, 9].

$$min \sum_{(k,i,j,t)} TC_{(k,i,j,t)} \tag{1}$$

3. Implementation in the case study

Initially, the models were run in three cases comprising a standalone version of the UK and NL+SM networks and an integrated (UK+NL+SM) scenario to set up the base cases (i.e. 0, 1, 2 cases) to be compared against the what-if scenarios. Fig. 2 illustrates the boundary of these networks, their routing for batchwise and offshore pipeline transport options, and the stationary emitters considered within each. The UK concept is assumed to be formed by the Net Zero Teesside (NTZ) and Solent area stationary emitters, representing a cluster of individual stationary emitters connected to their intermediate storage Hub (Teesside hub and Solent Hub). The NL+SM concept combines individual stationary emitters and clusters of emitters. The stationary emitters from NL are Yara (1), a fertiliser plant at Herøya in Porsgrunn, Fortum Oslo Varme -FOV (3), a waste incineration plant in Klemetsrud, near Oslo, Preem (4), a refinery in Lysekil, Sweden.

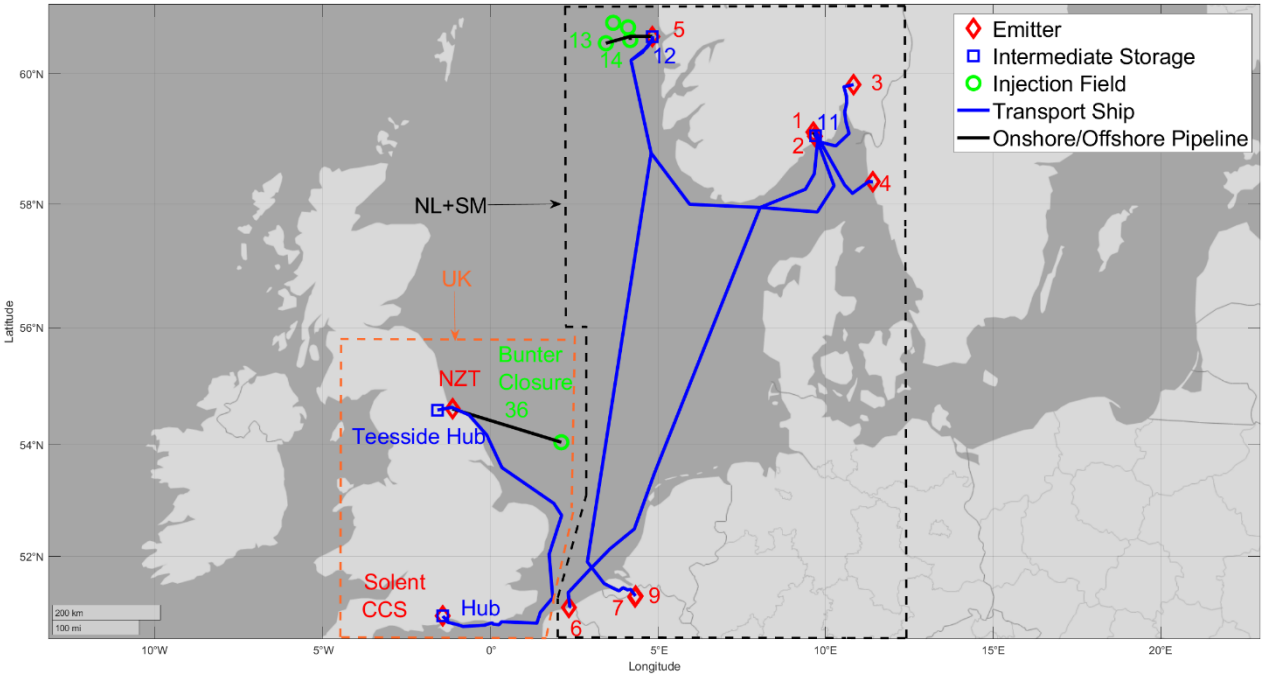


Fig. 2. Overview of CO_2 sources (1–9; Solent CCS, NZT), intermediate storage site hubs (10,11, Solent Hub, Teeside Hub), and injection fields (13–14) considered in this study – Yara (1), Norcem (2), Fortum Oslo Varme FOV (3), Preem (4), Kollsnes (5), Dunkirk (6), Antwerp (7), SL-F1 (9), Solent CCS, NZT, Hub-Grenland (10), Hub-Kollsnes (11), Aurora field (13) Smeaheia γ field (14). Ship and offshore pipeline routes.

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The clusters of emitters are Norcem (2), a cement plant in Brevik, which is assumed to collect CO_2 from other stationary emitters nearby, and Kollsnes (5), which is expected to receive CO_2 from other networks and locally. This concept uses two intermediate storage hubs, Hub-Grenland (10) in Grenland, and Hub-Kollsnes (11), to establish a transport route of CO_2 by ship towards the storage sites Aurora(13), Smeaheia γ (14), which are connected to Hub-Kollsnes (11), via offshore pipeline [3]. The Stella Maris concept contains three ports concentrating the CO_2 captured in the surrounding industry clusters. In France, the Dunkirk (6) port is considering the evolution of the 3D project. In Belgium, the Antwerp port (7) is a source of CO_2 captured from multiple chemical processing plants and also has SL-F1 (9), a hypothetical small-scale emitter intended to incorporate small-scale CO_2 transported by lorries.

The aggregated emission profile from the UK was assumed to be 2,016 kt CO_2 in the first year and to reach 2,034 kt CO_2 in the 30th year. For NL+SM, the aggregated emission profile was assumed to be 1,001in the first year and to reach 28,402 kt CO_2 in the last year. The aggregate emission in both networks was assumed to be 333.8 Mt CO_2 for the whole time horizon. Fig. 3 illustrates the emission profile per source considered in the study, thus forming the base case emission profile.

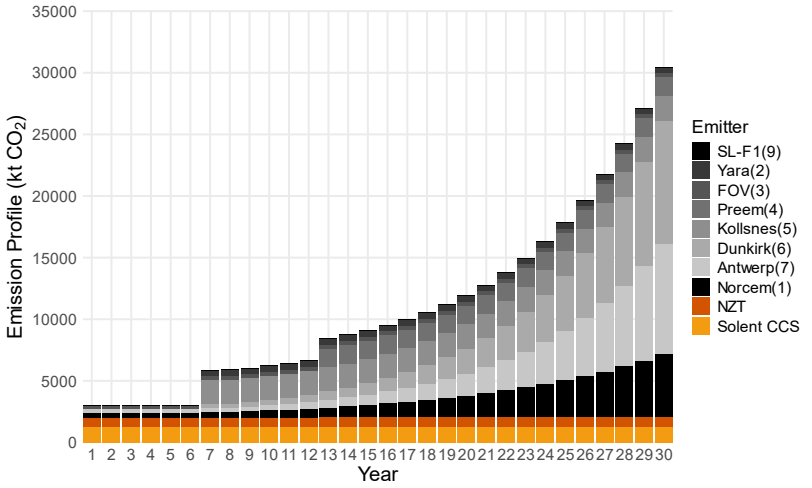


Fig. 3. Emission profile for the base case with emitters (1–9), in grey, forming the NL+SM, Solent CCS, and NZT, in orange, forming the UK.

The what-if scenarios were conceived with four main drivers: emission profile growth (i.e. scenarios a, b, c and d), delay in stationary emitter connection to the system (i.e. scenarios e, f, g and h), delay in storage site availability (i.e. scenarios y and z), and interconnection between the networks (i.e. cases x and i). Table 1 summarises the what-if scenarios devised, emphasising which element has been modified for the analysis.

Table 1. CCS network what-if scenarios conceptualisation.

Scenario/	**Integrated**	**Standalone**	
Case	**UK+NL+SM**	**UK**	**SM**
Base case	**0**	**1**	**2**
Higher emission profile 2% a.a. growth	**a)** UK **c)** NL+SM	**b)**	**d)**
Delay in emission profile (6 years)	**e)** Solent **g)** Kolsness and Norcem growth	**f)** Solent	**h)** Kolsness and Norcem growth
Delay in storage site availability (6 years)	**y)** Aurora	n/a	n/a
Interconnection between scenarios	**x)** NL+SM available to the UK after 10 years **i)** UK storage site available to SM	n/a	n/a

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4. Results

The 15 what-if scenarios are systematically evaluated against their corresponding base case to evaluate the batchwise transport opportunity in terms of key metrics, namely total cost, unit cost, levelised cost (i.e. discounted at 10%), emission profile, ship fleet size, ship trips per year, and the accumulated quantity of wells in the network. Table 2 summarises the results and compares the cost metrics of all scenarios relative to the respective base case. Summarising the results for all scenarios against their respective base case, the yearly percentage change in CO_2 emission varied by ±62%, resulting in a yearly variation of total optimal cost ranging from -71% to 316%, a yearly ship fleet (i.e., considering all ship capacities) variation from -3 to 5 ships, and yearly trips variation from -212 to 343 trips. A comparison throughout the the 30-year time horizon for the network design metrics (i.e. emission profile, total cost, and batchwise network) for each scenario is presented in the appendix. Furthermore, Fig. 4. illustrates the cost metrics across all cases and scenarios and a cross-scenario heat map with the variance against their respective base case.

Table 2. Results comparison of cost metrics between what-if scenarios and their respective base case.

Context	Base case	Scenario	Scenario performance in relation to the respective base case
Higher emission profile (2% a.a. growth)	Integrated (UK+NL+SM)	**a)** UK **c)** NL+SM	**Higher total costs and mostly lower levelised and unit costs.** Total cost increase range: 6.4% to 37.1% Levelised cost variation range: -4.8% to -4.8% Unit cost variation range: 5.8% to 1.3%
	UK standalone	**b)**	
	NL+SM standalone	**d)**	
Delay in emission Profile (6 years)	Integrated (UK+NL+SM)	**e)** UK **g)** NL+SM	**Higher levelised and unit costs. Reduced total cost if the infrastructure can be adapted.** Total cost variation range: -1.3% to 0.6% Levelised cost increase range: 1.4% to 9.8% Unit cost increase range: 7.1% to 27.9%
	UK standalone	**f)**	
	NL+SM standalone	**h)**	
Delay in storage site availability (6 years Aurora)	Integrated (UK+NL+SM)	**y)**	**Higher total, levelised and unit costs. Lower levelised cost if interconnection is possible.** Total cost variation range: 7.7% to 7.8% Levelised cost increase range: 7.7% to 7.8% Unit cost increase range: -0.4% to 4.6%
	NL+SM standalone	**z)**	
Interconnection between cases	Integrated (UK+NL+SM)	**x)** UK to NL **i)** SM to UK	**Lower total, levelised and unit costs. Increased levelised cost for scenario x.** Total and unit cost reduction: -0.3% Levelised cost increase range: -0.5% to 3.3%

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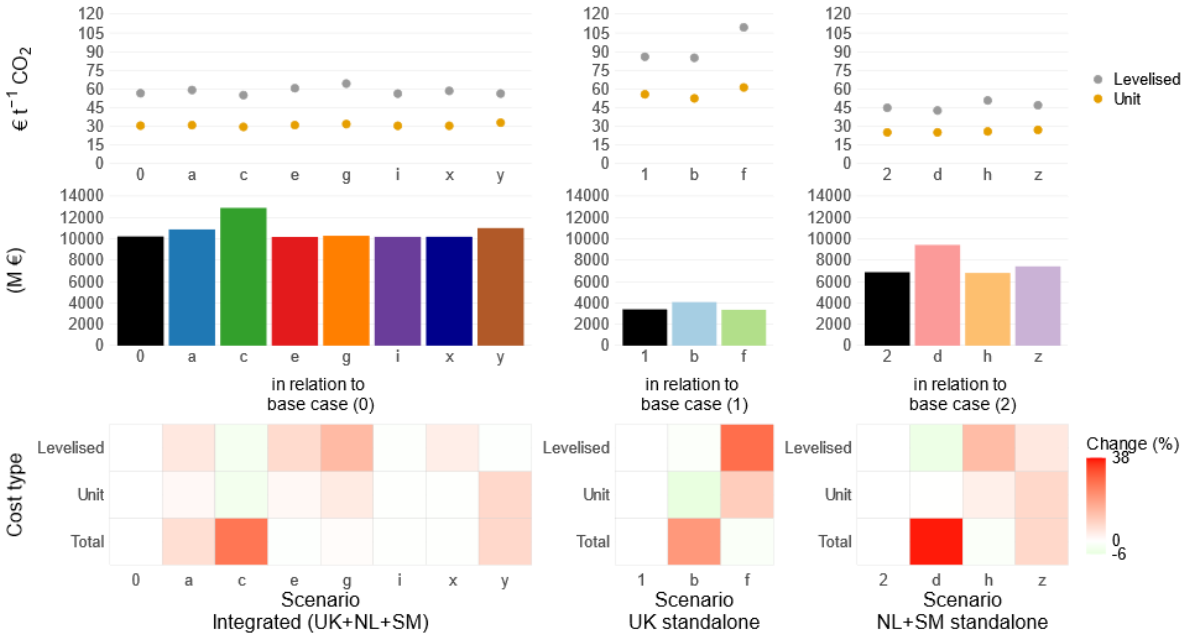


Fig. 4. Overview of what-if scenario cost metrics and performance against their respective base case.

5. Conclusions

This study has developed and analysed three distinct batchwise CCS network cases — standalone and integrated — and a total of 15 what-if scenarios designed to assess the value that CO_2 shipping flexibility can offer in evolving CCS networks. The integrated scenario encompassing the UK, Northern Lights (NL), and Stella Maris (SM) concepts demonstrated reduced variability across all cost metrics analysed, including total cost, levelised cost, and unit cost, as compared to the standalone configuration across most cases for the design parameters adopted in the analysis. These results indicate the positive economic advantages of a coordinated CCS network infrastructure.

Moreover, the integrated UK+NL+SM scenario achieved a lower optimal ship fleet size, suggesting that fleet sharing and trip optimisation can be viable strategies for reducing operational costs and improving logistical efficiency (i.e. base cases). Allowing the interconnection (i.e. SM to the UK) marginally lowered the cost metrics, indicating potential synergies from ship trips. The opposite, i.e. considering that the UK may direct CO_2 towards NL after 10 years, has not changed the network configuration, indicating that once storage sites become operational, with the relevant infrastructure in place (i.e. offshore pipelines), the cost differentials between storage sites must be material to change the economics and divert the CO_2 flow.

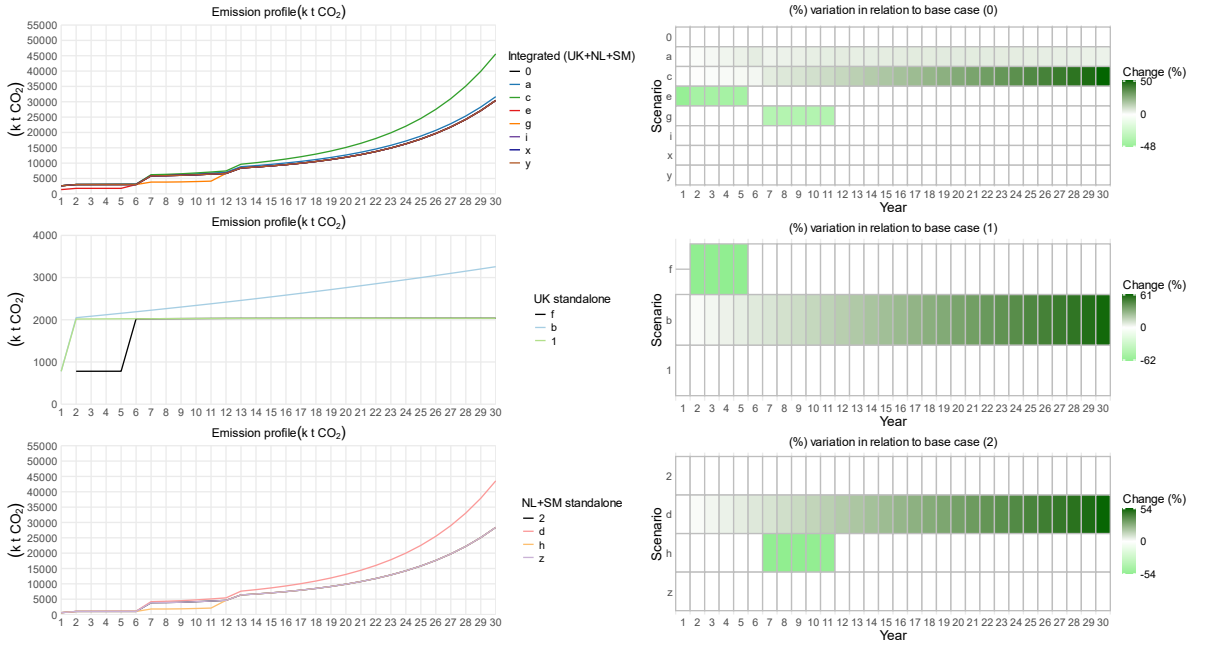
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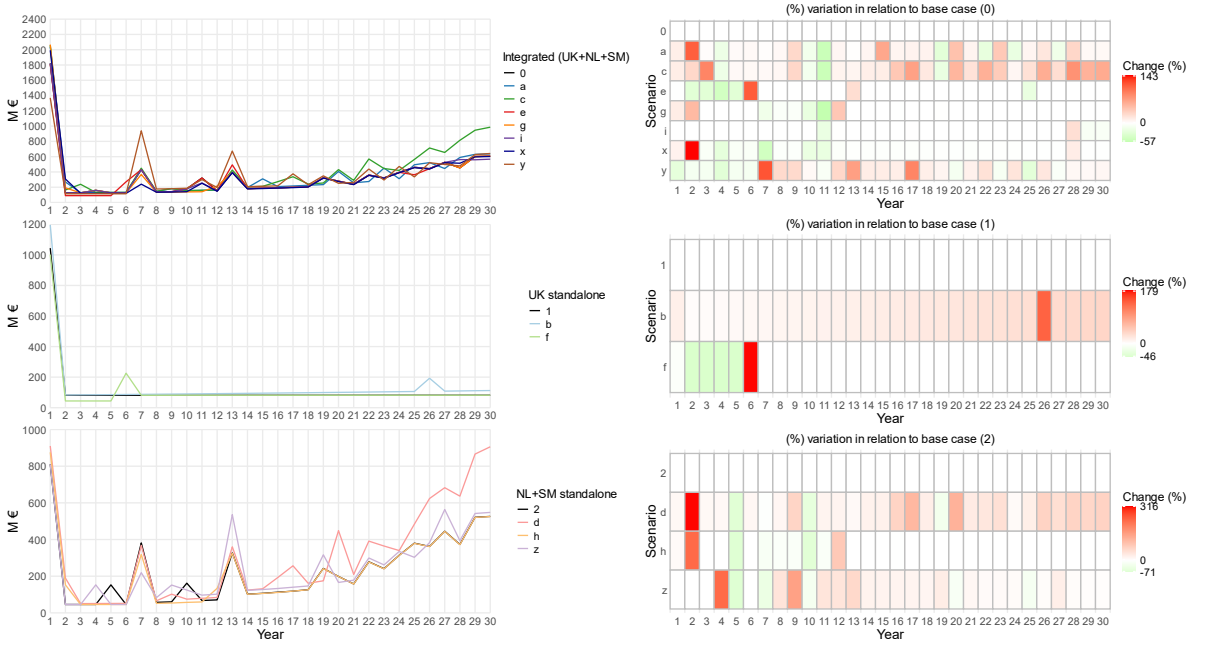
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Appendix A. Detailed metrics for the what-if-scenarios against their respective base case

A.1. Emission profile

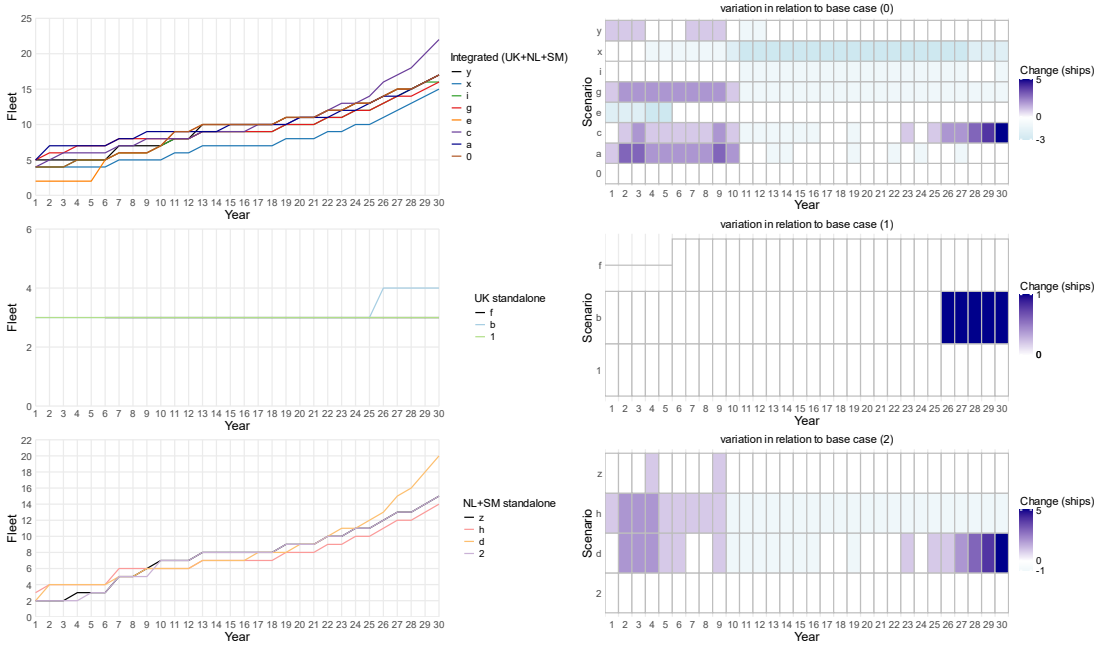


A.2. Total Cost

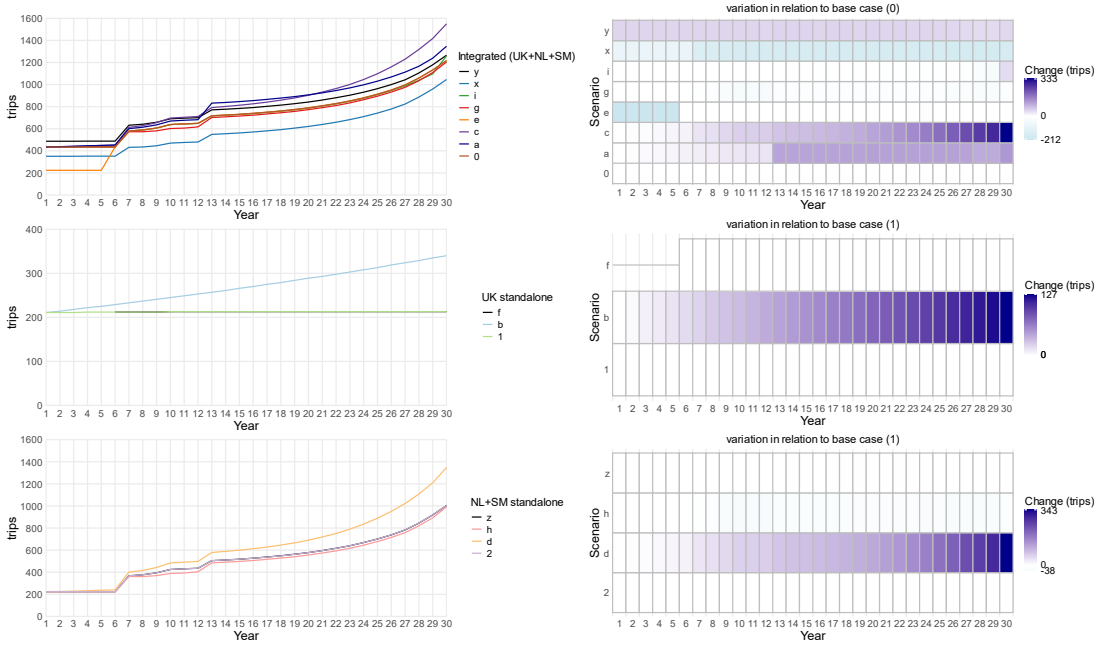


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A.3. Ship Fleet



A.4. Ship Trips



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